

Rohit Dhakal

Full-Stack Developer — Backend-Focused Engineer

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EDUCATION

Galgotias University

Bachelor of Technology in Computer Science and Engineering

Uttar Pradesh, India

2022 – 2026

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, SQL

Frontend: React.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Databases & Caching: MongoDB, Redis, Firebase Firestore

Core Concepts: Data Structures & Algorithms, OOP, MVC Architecture, Cache-Aside Pattern, WebSocket Communication, System Design Basics

Tools: Git, GitHub, Postman, Bruno, Vercel, Render

PROJECTS

AI-Powered Health Platform — TypeScript, MongoDB, Express, React, Node.js, Gemini API

- Engineered AI-driven nutrition parser using Gemini API to convert natural language inputs into structured JSON for calorie and exercise tracking
- Designed scalable MVC-based backend architecture with centralized error handling and request validation middleware
- Developed type-safe REST APIs with JWT-based authentication and protected route handling for secure user sessions
- Optimized request-response flow to ensure consistent API performance under concurrent usage
- Deployed full-stack application on Vercel and Render with environment-based configuration management
- Live:** ai-health-tracker-zg2o.vercel.app — **GitHub:** github.com/Rohitdhakal1/Ai-Health-Tracker

URL Shortener with Redis Caching (ShrinkIt) — Node.js, TypeScript, Redis, MongoDB

- Implemented Cache-Aside pattern using Redis to reduce database read operations and improve API response latency
- Designed scalable RESTful API supporting URL shortening, validation, and click analytics tracking
- Built fault-tolerant fallback mechanism to MongoDB ensuring high availability during cache misses
- Developed custom short URL generation logic with collision detection and handling
- Structured service layer separation for maintainable and extensible backend architecture
- Live:** shortenit-frontend-jwxs.vercel.app — **GitHub:** github.com/Rohitdhakal1/ShrinkIt-url-shortner

Comfi_Space – Real-Time Chat Application — React, TypeScript, Socket.IO, Express, Node.js

- Built real-time chat system using Socket.IO enabling bi-directional WebSocket communication
- Handled concurrent client connections using event-driven architecture for real-time message broadcasting
- Implemented room-based messaging, user presence tracking, and timestamp synchronization
- Designed responsive chat interface using React and Tailwind CSS
- Managed in-memory session storage to optimize real-time performance without persistent database overhead
- GitHub:** github.com/Rohitdhakal1/Comfi_Space-Chat-using-socket-

AI Presentation Generator — React, Firebase, Gemini API, Clerk Auth

- Developed AI-powered slide generation system using Gemini API to transform user prompts into structured presentation data
- Designed dynamic slide rendering engine and implemented PDF export pipeline for downloadable presentations
- Integrated Firebase Firestore for real-time synchronization and Clerk authentication for secure user management
- Built modular and reusable UI components with live preview and customizable templates

ACHIEVEMENTS

Solved **189+ LeetCode problems** with focus on Trees, Dynamic Programming, and Medium-level algorithmic challenges

Built and deployed **5+ full-stack MERN applications** integrating AI services, authentication systems, and caching mechanisms